# ClipSAT : Math for AI & Data Science [ANSWER KEY]

## Unit 5.01: Distributions & Maximum Likelihood

**Q1.** A feature in an AI vision model follows a normal distribution X sim mathcal{N}25, 36 . What is the mean mu and variance sigma^2 of this feature?

**[x] A: mu = 25, ; sigma^2 = 36**
[ ] B: mu = 36, ; sigma^2 = 25
[ ] C: mu = 25, ; sigma^2 = 6
[ ] D: mu = 6, ; sigma^2 = 36

Solution: The notation mathcal{N}mu, sigma^2 directly specifies the mean mu = 25 and variance sigma^2 = 36 with standard deviation sigma = 6 .

**Q2.** Given n = 4 independent observations from a normal distribution with sum sum_{i=1}^{4} x_i = 12 , what is the Maximum Likelihood Estimate MLE hat{mu}_{text{MLE}} ?

**[x] A: hat{mu}_{text{MLE}} = 3**
[ ] B: hat{mu}_{text{MLE}} = 6
[ ] C: hat{mu}_{text{MLE}} = 12
[ ] D: hat{mu}_{text{MLE}} = 1.5

Solution: The log-likelihood ln Lmu = -frac{n}{2}ln2pisigma^2 - frac{1}{2sigma^2}sum x_i - mu^2 . Setting frac{partial ln L}{partial mu} = 0 yields the sample mean hat{mu} = frac{1}{n}sum x_i = frac{12}{4} = 3 .

**Q3.** Given n = 8 independent observations from a normal distribution with sum sum_{i=1}^{8} x_i = 24 , what is the Maximum Likelihood Estimate MLE hat{mu}_{text{MLE}} ?

**[x] A: hat{mu}_{text{MLE}} = 3**
[ ] B: hat{mu}_{text{MLE}} = 6
[ ] C: hat{mu}_{text{MLE}} = 24
[ ] D: hat{mu}_{text{MLE}} = 1.5

Solution: The log-likelihood ln Lmu = -frac{n}{2}ln2pisigma^2 - frac{1}{2sigma^2}sum x_i - mu^2 . Setting frac{partial ln L}{partial mu} = 0 yields the sample mean hat{mu} = frac{1}{n}sum x_i = frac{24}{8} = 3 .

**Q4.** Given n = 12 independent observations from a normal distribution with sum sum_{i=1}^{12} x_i = 36 , what is the Maximum Likelihood Estimate MLE hat{mu}_{text{MLE}} ?

**[x] A: hat{mu}_{text{MLE}} = 3**
[ ] B: hat{mu}_{text{MLE}} = 6
[ ] C: hat{mu}_{text{MLE}} = 36
[ ] D: hat{mu}_{text{MLE}} = 1.5

Solution: The log-likelihood ln Lmu = -frac{n}{2}ln2pisigma^2 - frac{1}{2sigma^2}sum x_i - mu^2 . Setting frac{partial ln L}{partial mu} = 0 yields the sample mean hat{mu} = frac{1}{n}sum x_i = frac{36}{12} = 3 .

**Q5.** Given n = 16 independent observations from a normal distribution with sum sum_{i=1}^{16} x_i = 48 , what is the Maximum Likelihood Estimate MLE hat{mu}_{text{MLE}} ?

**[x] A: hat{mu}_{text{MLE}} = 3**
[ ] B: hat{mu}_{text{MLE}} = 6
[ ] C: hat{mu}_{text{MLE}} = 48
[ ] D: hat{mu}_{text{MLE}} = 1.5

Solution: The log-likelihood ln Lmu = -frac{n}{2}ln2pisigma^2 - frac{1}{2sigma^2}sum x_i - mu^2 . Setting frac{partial ln L}{partial mu} = 0 yields the sample mean hat{mu} = frac{1}{n}sum x_i = frac{48}{16} = 3 .

## Unit 5.01: Distributions & Maximum Likelihood

**Q6.** Given n = 20 independent observations from a normal distribution with sum sum_{i=1}^{20} x_i = 60 , what is the Maximum Likelihood Estimate MLE hat{mu}_{text{MLE}} ?

**[x] A: hat{mu}_{text{MLE}} = 3**
[ ] B: hat{mu}_{text{MLE}} = 6
[ ] C: hat{mu}_{text{MLE}} = 60
[ ] D: hat{mu}_{text{MLE}} = 1.5

Solution: The log-likelihood ln Lmu = -frac{n}{2}ln2pisigma^2 - frac{1}{2sigma^2}sum x_i - mu^2 . Setting frac{partial ln L}{partial mu} = 0 yields the sample mean hat{mu} = frac{1}{n}sum x_i = frac{60}{20} = 3 .

**Q7.** Given n = 24 independent observations from a normal distribution with sum sum_{i=1}^{24} x_i = 72 , what is the Maximum Likelihood Estimate MLE hat{mu}_{text{MLE}} ?

**[x] A: hat{mu}_{text{MLE}} = 3**
[ ] B: hat{mu}_{text{MLE}} = 6
[ ] C: hat{mu}_{text{MLE}} = 72
[ ] D: hat{mu}_{text{MLE}} = 1.5

Solution: The log-likelihood ln Lmu = -frac{n}{2}ln2pisigma^2 - frac{1}{2sigma^2}sum x_i - mu^2 . Setting frac{partial ln L}{partial mu} = 0 yields the sample mean hat{mu} = frac{1}{n}sum x_i = frac{72}{24} = 3 .

**Q8.** In training a language model token predictor, a binary token occurs k = 3 times across n = 10 independent trials. What is the Maximum Likelihood Estimate hat{p}_{text{MLE}} of the token probability?

**[x] A: hat{p}_{text{MLE}} = 0.300**
[ ] B: hat{p}_{text{MLE}} = 0.450
[ ] C: hat{p}_{text{MLE}} = 0.150
[ ] D: hat{p}_{text{MLE}} = 0.700

Solution: For a Bernoulli likelihood Lp = p^k 1-p^{n-k} , the log-likelihood is ln L = kln p + n-kln1-p . Setting derivative to zero: frac{k}{p} - frac{n-k}{1-p} = 0 implies hat{p} = frac{k}{n} = frac{3}{10} = 0.300 .